AI-ASSISTED ALERT TRIAGE IN SECURITY OPERATIONS CENTERS

Minor Project Report

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Abstract

Security Operations Centers (SOCs) monitor organizational systems, investigate security alerts and coordinate responses to potential cyber threats. However, high alert volumes, false positives and limited analyst resources can make it difficult to identify and prioritize genuine threats efficiently. This project examines how artificial intelligence (AI) can support alert triage through alert classification, prioritisation and explainable decision support. The study uses descriptive qualitative secondary research, drawing on academic publications, cybersecurity frameworks and technical documentation. It reviews alert fatigue, machine-learning approaches to security analysis, human-AI collaboration and explainable AI. The reviewed literature indicates that AI may help analysts organize and assess alerts, but its usefulness depends on data quality, model reliability, task-appropriate evaluation and human oversight. Classification and prioritisation are distinct tasks, and strong performance in one does not guarantee strong performance in the other. Explainability can make model outputs easier to inspect, although explanations do not prove that an alert is malicious. The project recommends a human-centred approach in which AI provides recommendations and contextual information while analysts validate evidence and retain responsibility for consequential decisions. Alongside the literature review, a synthetic Python prototype compared a context-aware rule engine with a categorical Naive Bayes classifier on a 35-alert holdout. Both achieved 88.6% accuracy; macro-F1 was 0.897 for the rule engine and 0.885 for Naive Bayes. These results illustrate an evaluation workflow and do not establish performance in an operational SOC. The study synthesizes published findings and develops practical recommendations.

1. Introduction

1.1 Background

A Security Operations Center (SOC) is a team or organizational function responsible for monitoring information systems, identifying suspicious activity, investigating potential incidents and coordinating defensive responses. SOC analysts use security information and event management platforms, endpoint detection tools, network monitoring systems and threat intelligence to identify possible threats. These technologies generate alerts when activity matches a detection rule, deviates from an established pattern or appears to indicate malicious behaviour.

Not every alert represents a genuine incident. Legitimate activity can trigger detection rules and produce false positives. Excessive volumes can contribute to alert fatigue, making important events harder to recognize. Tariq et al. (2025) review its causes and mitigation approaches, including automation, augmentation and human-AI collaboration.

AI and machine learning may help analyse alert characteristics, identify patterns and classify or prioritise events. However, outputs are not automatically reliable. Rieger et al. (2026) report that large language model performance differs between alert classification and prioritisation.

Explainability is also important. Analysts should be able to inspect why an AI system assigned an alert a particular priority. SHAP and LIME are among the methods reviewed in explainable intrusion-detection research (Mohale & Obagbuwa, 2025). Explanations support investigation but do not replace verification.

1.2 Research Question

This project is guided by the following research question: How can AI-assisted alert triage support SOC efficiency while addressing false positives, explainability and the need for human oversight?

2. Objectives of the Study

This project examines how AI can support SOC alert triage while maintaining accuracy, transparency and human oversight. Its specific objectives are:

- To examine the operational challenges faced by SOC analysts during alert triage, particularly high alert volumes, false positives and alert fatigue.
- To study how AI and machine-learning techniques can support the classification and prioritisation of security alerts.
- To analyse the benefits and limitations of AI-assisted triage, including model reliability, data quality, false positives, missed threats and explainability.
- To examine the role of human-AI collaboration in ensuring that automated recommendations are reviewed and validated by SOC analysts.
- To recommend a practical, human-centred approach for integrating AI-assisted alert triage into SOC workflows.

3. Literature Review / Background Study

3.1 Security Operations Centers and Alert Fatigue

SOCs depend on alerts generated by security tools to identify suspicious activity. However, a large volume of alerts can make it difficult to distinguish genuine threats from harmless or irrelevant events. Tariq et al. (2025) identify alert fatigue as an important SOC research and operational challenge. Their review discusses contributing factors such as alert volume, false alarms, staffing constraints, fragmented tools and insufficient context. Repeatedly investigating alerts that do not represent genuine threats can divert time and attention from potentially serious incidents.

Alert quality, relevance and prioritisation matter as much as volume. Triage should identify events requiring immediate investigation while avoiding the filtering of genuine threats.

3.2 AI-Assisted Alert Classification and Prioritisation

AI and machine learning can support alert triage by analysing alert information and identifying patterns that may indicate suspicious activity. Classification assigns an alert to a category, such as potentially malicious or benign. Prioritisation determines how urgently the alert should be investigated. Rieger et al. (2026) study large language models for both tasks and compare their performance with traditional machine-learning approaches. Their findings indicate that several models achieved strong recall in binary alert classification, while prioritisation performance was weaker and could produce noisy recommendations.

Identifying a potentially malicious alert and deciding its urgency are different tasks. A model may classify correctly but assign an unsuitable priority, so each task requires separate evaluation.

3.3 Machine Learning in Intrusion Detection

Machine-learning techniques provide additional background for understanding automated security analysis. Ur Rehman et al. (2025) review supervised, unsupervised, deep-learning and hybrid approaches used in intrusion detection systems. Their review discusses challenges involving dataset quality, class imbalance, generalisation, explainability and the difficulty of comparing results across different datasets. These limitations are relevant when considering whether a model trained on one dataset will perform reliably in a different operational environment.

Intrusion detection and SOC alert triage are related but distinct activities. Intrusion-detection research can inform the technical background of this project, but its findings should not

automatically be treated as direct evidence that a particular model will improve SOC triage performance.

3.4 Human-AI Collaboration in Security Operations

Fully automated decision-making creates risks when model outputs are incorrect or lack sufficient context. Security alerts may require investigation using information about affected assets, user activity, network behaviour and threat intelligence. Chhetri et al. (2024) present a conceptual framework for human-AI teaming in SOCs. Their A²C framework supports transitions between automated, augmented and collaborative modes of decision-making. It is a proposed framework and research vision, not a controlled deployment study proving operational effectiveness.

The National Institute of Standards and Technology's AI Risk Management Framework provides a broader foundation for responsible AI use. Tabassi (2023) presents four core functions: Govern, Map, Measure and Manage. The framework is not specific to SOC alert triage, but its principles help organizations consider oversight responsibilities, operational context, system evaluation and ongoing risk management.

3.5 Explainability and Trust in AI-Assisted Detection

A major challenge with some machine-learning models is that their predictions may be difficult for humans to interpret. Mohale and Obagbuwa (2025) systematically review explainable AI approaches used in intrusion detection systems, including SHAP, LIME, rule-based approaches and hybrid methods. Their review discusses the value of explanations for transparency while identifying challenges such as computational overhead, scalability, standardisation and trade-offs between interpretability and predictive performance.

An explanation may show that repeated failed logins or unusual network activity influenced a suspicious classification. It describes model behaviour; it does not prove that an event is malicious.

3.6 Detection Rules and Cybersecurity Context

AI-assisted triage should operate alongside existing security processes. Elastic's documentation describes detection rules, event correlation, threshold-based detection and machine-learning anomaly detection, as well as validating and tuning rules to reduce noise. MITRE ATT&CK provides a knowledge base of adversary tactics and techniques that can help defenders describe observed behaviour and support investigation. A mapping to an ATT&CK technique does not automatically confirm malicious activity.

Overall, the literature suggests that AI can contribute to alert classification, prioritisation and decision support, but its usefulness depends on reliable data, appropriate evaluation, explainability and human oversight. These findings provide the foundation for the analysis and recommendations in this project.

4. Research Methodology

4.1 Research Design

This project adopts a descriptive and qualitative secondary-research approach. It examines existing academic publications, cybersecurity frameworks and technical documentation related to AI-assisted alert triage, intrusion detection, alert fatigue and explainable artificial intelligence. Secondary research was selected because the project aims to understand existing approaches, compare their reported benefits and limitations, and develop recommendations for SOC

environments. No primary data was collected from organizations, SOC analysts or operational security systems.

4.2 Sources of Information

The study uses peer-reviewed research papers and systematic reviews, official cybersecurity and AI frameworks, and technical documentation. The selected sources cover alert fatigue and SOC operations; AI and machine-learning approaches to classification and prioritisation; human-AI collaboration; explainability; and security detection and alert-tuning practices. Academic publications provide research findings, while NIST, MITRE and Elastic provide framework or technical context.

4.3 Source Selection and Analysis

Sources were selected for relevance, credibility and contribution to the objectives, with emphasis on peer-reviewed research and official documentation. Intrusion-detection studies provide technical context but are distinguished from research specifically about SOC workflows.

The material was analysed thematically across alert fatigue, classification, prioritisation, explainability, human oversight and implementation. Findings were compared to identify common themes and limitations. Classification performance does not establish prioritisation quality, and an explanation does not confirm an attack.

4.4 Scope and Limitations

The study focuses on AI-assisted alert triage within SOC environments rather than every application of AI in cybersecurity. Some reviewed studies focus on intrusion detection or explainable AI rather than SOC workflows directly. Their findings are therefore used as supporting technical context and not as direct proof of improved SOC performance. The project relies on published research and documentation, so its conclusions are limited by the methods,

datasets and findings reported by the original authors. Differences in datasets, evaluation measures and operational environments can also make direct comparisons difficult. This research does not independently validate the effectiveness of a commercial AI triage platform.

4.5 Supporting Practical Demonstration

A supplementary Python prototype was implemented using fictional security alerts. The original rule-only dataset contains 64 alerts. A separate context-rich dataset contains 98 development and training alerts, with 14 examples for each of seven event types, and a fresh holdout contains 35 alerts, with five examples per event type. Ground-truth classes were assigned from the scenario narratives and structured evidence independently of the model predictions. The prototype compares a context-aware rule engine with a categorical Naive Bayes classifier. The classifier uses structured event and context fields; it excludes alert identifiers, source IP addresses, descriptions, severity, and ground-truth labels. The numeric failed-attempt and transfer fields are grouped into fixed ranges. All data are synthetic, and the demonstration does not use organizational logs or represent a production SOC system.

5. Analysis and Discussion

5.1 Alert Volume and Analyst Workload

The reviewed literature indicates that alert fatigue is not simply a technical problem caused by security tools generating too many notifications. It is also an operational challenge involving alert quality, analyst capacity and investigation workflows. Tariq et al. (2025) discuss how high alert volumes, false positives and operational constraints can make it difficult for SOC analysts to identify important events. Repeatedly investigating alerts that do not represent genuine threats can divert time and attention from potentially serious incidents.

An effective triage process should not focus only on reducing the number of alerts. It should improve the relevance and organisation of the information presented to analysts. AI-assisted triage may contribute by helping classify alerts, identify patterns and organise events for further investigation. However, reducing alert volume without considering missed threats could create another problem: important events might be incorrectly filtered or assigned low priority. Workload reduction must therefore be balanced with the ability to detect and investigate genuine threats.

5.2 Classification and Prioritisation Are Different Challenges

Classification attempts to assign an alert to a category, such as potentially malicious or benign. Prioritisation determines how urgently it should be investigated. Two alerts may both be suspicious, but one may involve a critical server while the other concerns a lower-risk asset. Rieger et al. (2026) highlight that performance can differ between classification and prioritisation. Strong performance in identifying true and false alerts does not automatically mean that a model can reliably assign priority levels.

A triage system should evaluate both tasks separately and consider asset importance, affected accounts, event relationships and threat intelligence. Evaluation should include precision, recall and F1-score, not accuracy alone: false positives increase workload, while false negatives can leave threats undetected.

5.3 Role and Limitations of Machine Learning

Machine-learning techniques can identify patterns in security data and support the detection or classification of suspicious activity. Ur Rehman et al. (2025) review several families of machine-learning methods used in intrusion detection. The reviewed research also identifies challenges involving data quality, class imbalance, generalisation and explainability. A model trained on

one dataset may perform differently when exposed to new attack patterns or a different network environment. If benign events greatly outnumber malicious events, a model may achieve high accuracy while failing to identify a meaningful proportion of actual threats.

Machine learning is not a universal solution to alert fatigue. Models require relevant evaluation data and suitable performance measures before their outputs inform investigation priorities.

5.4 Importance of Human Oversight

The reviewed literature supports an approach in which AI assists analysts rather than completely replacing their judgement. Chhetri et al. (2024) discuss human-AI teaming as a way to combine automated support with human decision-making in SOC environments. AI can help process information and present recommendations, but analysts may need context that is not available to the model. An unusual login, for example, may be suspicious in one situation but legitimate during approved maintenance.

The NIST AI Risk Management Framework provides a structure for considering these risks through Govern, Map, Measure and Manage (Tabassi, 2023). Applied to alert triage, these functions encourage organizations to establish oversight responsibilities, understand the system's context, evaluate performance and manage identified risks. AI recommendations should support analysts rather than be treated as confirmed incident decisions. Human oversight is especially important for high-impact actions such as isolating systems or disabling accounts.

5.5 Explainability as a Decision-Support Requirement

An AI system may produce a score without clear reasoning. Mohale and Obagbuwa (2025) review SHAP and LIME, which can help identify features influencing a prediction and guide further analyst investigation.

However, an explanation is not proof that a prediction is correct. An explanation method may describe the model's behaviour without establishing that the underlying event is malicious. Explainability should be combined with evidence validation, contextual investigation and human review. It should also be presented in a way analysts can understand without unnecessary delay or excessive detail.

5.6 Integration with Existing Security Processes

AI-assisted triage should complement existing security monitoring capabilities rather than replace them. Detection rules, threshold-based detection, event correlation and machine-learning anomaly detection can provide different signals for investigation (Elastic, n.d.). MITRE ATT&CK can provide additional context by helping defenders relate observed behaviour to documented adversary tactics and techniques (MITRE, n.d.). Such mapping can support investigation, but it does not independently confirm malicious activity.

A practical workflow could combine several sources of information. Existing detection rules generate alerts, correlation connects related events, and AI assists with classification or prioritisation. The system can then present the alert, relevant context, a suggested priority and an explanation to a SOC analyst. This allows established detection mechanisms to remain part of the workflow while AI supports the organisation and interpretation of security information.

5.7 Synthetic Prototype Evaluation

The prototype comparison used 98 labelled alerts for model training and a separate 35-alert holdout for evaluation. The holdout was not used to fit the classifier or select its settings. The evaluator removed each reference label before generating a prediction. The rules and classifier were evaluated on the same holdout so their results can be compared on these scenarios.

5.7.1 Evaluation Method

The Naive Bayes model used categorical event and context features, including authorization, user confirmation, destination trust, process evidence, privilege approval, domain assessment and malware verdict. Failed-login counts were binned as 0-4, 5-9 or 10 or more; outbound transfer was binned as 0-99 MB, 100-499 MB or 500 MB or more. Alert IDs, source IPs, descriptions, original severity and reference labels were excluded from the model features. Precision, recall and F1 were calculated for each class; macro averages give each class equal weight.

5.7.2 Results

Both methods classified 31 of 35 holdout alerts correctly, for 88.6% accuracy. The rule engine had macro precision, recall and F1 of 0.897, 0.903 and 0.897. Naive Bayes had macro precision, recall and F1 of 0.884, 0.888 and 0.885. Naive Bayes had higher F1 for suspicious alerts (0.870 versus 0.833) and alerts needing review (0.897 versus 0.857), while the rule engine had higher F1 for benign alerts (1.000 versus 0.889). The results show a different error balance, not a gain in overall accuracy.

Table 1. Classification metrics on the 35-alert holdout

Model	Class	Precision	Recall	F1
Rules	Benign	1.000	1.000	1.000
Rules	Suspicious	0.769	0.909	0.833
Rules	Needs review	0.923	0.800	0.857
Rules	Macro average	0.897	0.903	0.897
Naive Bayes	Benign	0.889	0.889	0.889
Naive Bayes	Suspicious	0.833	0.909	0.870
Naive Bayes	Needs review	0.929	0.867	0.897

Naive Bayes	Macro average	0.884	0.888	0.885

Table 2. Confusion matrices on the 35-alert holdout

Model	Actual class	Predicted benign	Predicted suspicious	Predicted needs review
Rules	Benign	9	0	0
Rules	Suspicious	0	10	1
Rules	Needs review	0	3	12
Naive Bayes	Benign	8	0	1
Naive Bayes	Suspicious	1	10	0
Naive Bayes	Needs review	0	2	13

These results are descriptive of a small, hand-authored synthetic dataset. The training set contained 98 alerts and each holdout class had only 9 to 15 examples. Labels are scenario-based judgments, not adjudicated outcomes from a SOC. The results cannot establish real-world performance or be compared directly with the earlier 64-alert evaluation, which used a different dataset and lacked the same structured context.

5.8 Overall Findings

The analysis identifies five findings: alert fatigue involves alert quality, volume and analyst capacity; classification and prioritisation require separate evaluation; data quality and generalisation affect model reliability; explainability does not guarantee correctness; and human oversight remains necessary. These findings support a human-centred model in which AI organizes information while analysts validate evidence and make consequential decisions.

6. Recommendations

6.1 Use AI to Support Analysts Rather Than Replace Them

AI should support repetitive classification and prioritisation, not independently make consequential security decisions. Analysts should validate alerts, investigate incidents and approve high-impact actions.

6.2 Evaluate Classification and Prioritisation Separately

Organizations should evaluate classification and prioritisation separately. Precision, recall, F1-score and confusion matrices should complement accuracy to reveal false positives and false negatives.

6.3 Maintain High-Quality and Relevant Data

Models should be evaluated on data representative of the operational environment. Organizations should consider data quality, class imbalance, outdated attack patterns and changes in network behaviour, and reassess models as conditions change.

6.4 Provide Understandable Explanations

Systems should provide understandable reasons for classifications and priorities. SHAP or LIME may identify influential features, but explanations are supporting information, not proof of an attack.

6.5 Integrate AI with Existing Security Tools

AI should complement existing security monitoring capabilities, including detection rules, event correlation and threat intelligence. Organizations should avoid depending entirely on a single model. Combining established detection mechanisms with AI-assisted analysis may provide a more practical approach to organising and investigating alerts.

6.6 Establish Governance and Continuous Monitoring

Organizations should assign responsibility for reviewing AI recommendations and monitoring performance. NIST's Govern, Map, Measure and Manage functions offer a risk-management structure. Ongoing evaluation should identify degradation and unexpected outputs.

6.7 Introduce AI-Assisted Triage Gradually

Organizations should initially test AI recommendations alongside existing SOC processes. Analysts should compare outputs with evidence before expanding the system's operational role.

7. Conclusion

This project examined how artificial intelligence can support alert triage in Security Operations Centers while maintaining accuracy, transparency and human oversight. The study was based on secondary research involving academic publications, cybersecurity frameworks and technical documentation related to alert fatigue, machine learning, explainable AI and security monitoring.

The findings indicate that high alert volumes, false positives and limited analyst resources can make security alert investigation challenging. AI-assisted classification and prioritisation offer potential ways to organise alerts and support analysts in identifying events that may require attention. However, classification and prioritisation are separate tasks, and a model's performance in one does not guarantee reliable performance in the other.

The study also identified important limitations involving data quality, class imbalance, model generalisation and explainability. AI-generated predictions should not be treated as automatically correct. Explainability techniques can help analysts understand the factors influencing model outputs, but explanations do not independently establish whether an event represents a genuine security incident.

The research further highlights the importance of human-AI collaboration. AI should assist with analysis and decision support, while SOC analysts retain responsibility for validating evidence and making consequential security decisions. Existing detection rules, event correlation and cybersecurity frameworks can complement AI-assisted approaches and provide additional operational context.

Overall, AI-assisted alert triage has the potential to support SOC operations when implemented carefully. Its practical usefulness depends on appropriate evaluation, relevant data, understandable outputs, continuous monitoring and human oversight. This project therefore recommends a human-centred approach in which AI enhances the capabilities of security analysts rather than replacing them.

The prototype adds a practical exercise to the secondary research, but its results are limited to hand-authored synthetic alerts. On a 35-alert holdout, the context-aware rules and categorical Naive Bayes model each classified 31 alerts correctly (88.6% accuracy); their macro-F1 scores were 0.897 and 0.885, respectively. These results do not establish performance in a real SOC. Future work could expand the dataset, obtain expert review of the labels, evaluate additional models, and use a new independent holdout for any further performance claims.

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